The Fisheye Models for Lens Distortion in 3DE4

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November 22, 2025

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1 About this document

1.1 License

The document may be shared and adapted under conditions of the Creative Commons license CC BY-SA 3.0 as described in [1].

1.2 Current status

The document is under development. It is not self-consistent, we are using notations and symbols from [2].

Vers.	Scope	Date	Notes
1.0	internal	2022-04-07	The fisheye mapping functions

2 The distortion models

2.1 Fisheye projection

Fisheye lenses are based on a projection function which maps any direction given in spherical coordinates to the rectangular filmback. There are four fundamental projection functions. While the fisheye projection functions already map in a non-linear way, we still have to tell this non-linearity apart from lens distortion. Even a perfect fisheye lens produces a "distorted" image, since the projection involves trigonometric functions. There is no such rule as "straight lines are mapped onto straight lines". On contrast, fisheye lenses have a different set of invariants, depending on the fisheye type.

In the following we denote coordinates in object space (= coordinate system of the camera) by $(x_{\text{obj}}, y_{\text{obj}}, z_{\text{obj}})$, where we assume that the coordinates are defined as usual in the post production industry, with y-axis pointing upwards and the camera looking towards $-z$. As usual we will assume that the camera is small in relation to the objects recorded, so that we can ignore depth of field effects. Any point in object space can then be described by spherical coordinates (ϕ, θ), which are related to $(x_{\text{obj}}, y_{\text{obj}}.z_{\text{obj}})$ by

$$\begin{aligned} x_{\text{obj}} &= \sin\theta\cos\phi \\ y_{\text{obj}} &= \sin\theta\sin\phi \\ z_{\text{obj}} &= -\cos\theta \end{aligned} \tag{1}$$

These coordinates project each point from three-dimensional space to a unit sphere around the camera by dropping the distance information. Point $(0, 0)$ in these coordinates coincides with the direction of view $(0, 0, -1)$.

Let us assume that the coordinates on the filmback are given by coordinates (x, y) the origin of which coincides with the axis of view. Let us further assume that there is no such thing like an anamorphic squeeze. Then we can describe a point on the filmback by two values (ϕ, r) with

$$\begin{aligned} x &= r\cos\phi \\ y &= r\sin\phi \end{aligned} \tag{2}$$

The fisheye projection is radially symmetric: A point (ϕ, θ) in object space is mapped onto a point (ϕ, r) on the filmback. Hence, we can restrict ourselves to the radial part of the projection. We shall call the radial part of the projection the *fisheye mapping* in accordance with the literature. The fisheye mappings are functions

$$F_{\text{any}} : [0, \alpha[\to \mathbb{R}^{\geq 0} \tag{3}$$

mapping 0 to 0, where α depends on the type of fisheye lens. Here, the subscript "any" stands for one of the four fisheye base types explained in the following sections. We shall denote the domain of the fisheye mapping function in $\mathbb{R}^2$ by

$$B_{\text{any}} \coloneqq \{p \in \mathbb{R}^2 | |p| < \alpha\} \tag{4}$$

The exact beam path for fisheye lenses is pretty complicated, but Figure 1 should give an idea of what we understand by "fisheye filmback space".

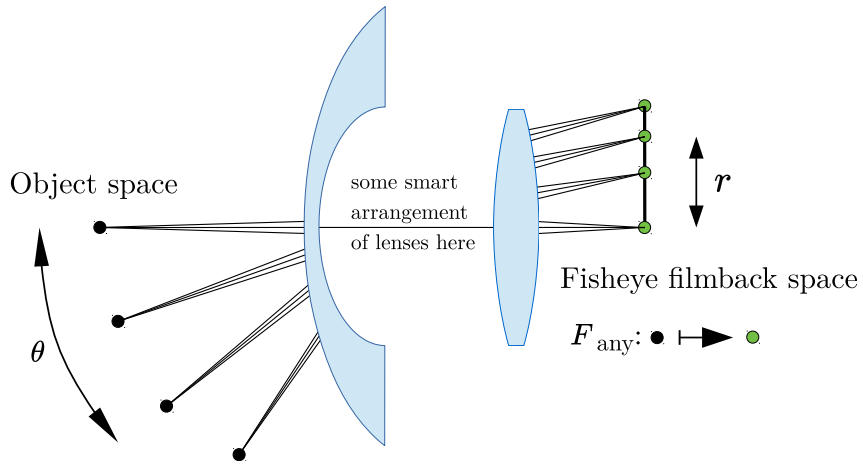


Figure 1: *Fisheye filmback space*

2.2 Remapping to plain filmback space

While the mapping function in the strict sense only establishes the relationship between the polar angle (with respect to z-axis) in object space and the distance between the corresponding point and the lens center on the fisheye filmback we need a complete mapping from filmback to filmback that removes the fisheye effect. The second part, the mapping from space to plain filmback space is the same for all models, see Figure 2. Again, we can restrict ourselves to the radial part. Let G be the radial part of the projection, then we have:

$$r'_{\text{dn}} = G(\theta) = f_{\text{dn}} \tan\theta \tag{5}$$

The full remapping then reads

$$r'_{\text{dn}} = G \circ F^{-1}_{\text{any}}(r_{\text{dn}}) \tag{6}$$

The derivative of this function according to the LEIBNIZ chain rule is

$$\frac{d}{dr_{\mathrm{dn}}} G \circ F_{\mathrm{any}}^{-1}(r_{\mathrm{dn}}) = \left.\frac{d}{d\theta}\right|_{F_{\mathrm{any}}(r_{\mathrm{dn}})} G(\theta) \left.\frac{d}{dr}\right|_{r_{\mathrm{dn}}} F_{\mathrm{any}}^{-1}(r) \tag{7}$$

and the derivative of G is:

$$\frac{d}{d\theta} G(\theta) = f_{\mathrm{dn}}(1 + \tan^2 \theta) \tag{8}$$

We will need this for computing the JACOBIAN which is required by our lens distortion classes.

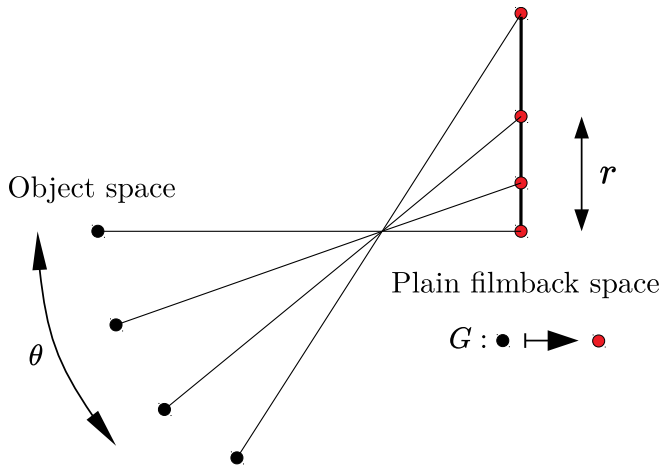


Figure 2: *Plain filmback space*

2.3 The projection types of fisheye lenses

In the following we shall have a look at the mapping function of each fisheye type. The diagrams illustrate the geometric relationship between the fisheye image (green dot) and the flattened, gnomonic image (red dot). The dot represents an object point projected by the fisheye lens or remapped to plain filmback space. The bold lines represent the radius of the projected point, that is green for r_{dn} and red for r'_{dn}.

2.3.1 The equidistant fisheye model

The mapping function for this lens type is

$$F_{\mathrm{ed}}(\theta) = f_{\mathrm{dn}}\theta \tag{9}$$

The derivative is

$$\frac{d}{d\theta} F_{\mathrm{ed}}(\theta) = f_{\mathrm{dn}} \tag{10}$$

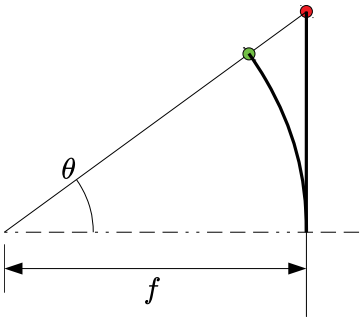


Figure 3: *Equidistant mapping*

The inverse function is

$$F_{\text{ed}}^{-1}(r_{\text{dn}}) = \frac{r_{\text{dn}}}{f_{\text{dn}}} \tag{11}$$

The derivative of the inverse function is

$$\frac{d}{dr_{\text{dn}}} F_{\text{ed}}^{-1}(r_{\text{dn}}) = \frac{1}{f_{\text{dn}}} \tag{12}$$

Line measures in radial direction are preserved under this fisheye mapping. Starting from a line integration measure $d\theta$ on the unit sphere we immediately get

$$dr = \frac{d}{d\theta} F_{\text{ed}}(\theta)\, d\theta = f_{\text{dn}}\, d\theta \tag{13}$$

which means the line measure is preserved up to a constant factor f_{dn} under the equidistant mapping.

2.3.2 The equisolid fisheye model

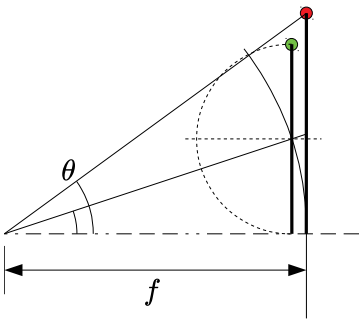


Figure 4: *Equisolid angle mapping*

The mapping function for this type of lens is

$$F_{\text{es}}(\theta) = 2 f_{\text{dn}} \sin \frac{\theta}{2} \tag{14}$$

The derivative is

$$\frac{d}{d\theta}F_{\text{es}}(\theta) = f_{\text{dn}} \cos\frac{\theta}{2} \tag{15}$$

The inverse function is

$$F_{\text{es}}^{-1}(r_{\text{dn}}) = 2 \arcsin\frac{r_{\text{dn}}}{2f_{\text{dn}}} \tag{16}$$

The derivative of the inverse function is

$$\frac{d}{dr_{\text{dn}}}F_{\text{es}}^{-1}(r_{\text{dn}}) = \frac{2}{\sqrt{(2f_{\text{dn}})^2 - {r_{\text{dn}}}^2}} \tag{17}$$

Any area measure is preserved under this fisheye mapping. In order to see this we map a surface element from the unit sphere to the filmback. Starting from the integration measure in filmback space

$$\begin{aligned} r dr\, d\phi &= 2f_{\text{dn}} \sin\frac{\theta}{2}\frac{d}{d\theta}F_{\text{es}}(\theta)\, d\theta\, d\phi \\ &= 2f_{\text{dn}}^2 \sin\frac{\theta}{2}\cos\frac{\theta}{2}\, d\theta\, d\phi \end{aligned} \tag{18}$$

we get after applying a common theorem for trigonometrics:

$$r dr\, d\phi = f_{\text{dn}}^2 \sin\theta\, d\theta\, d\phi \tag{19}$$

which is the solid angle integration measure in the unit sphere multiplied by a constant f_{dn}^2. In other words, the integration measure (and thus any area) is preserved under the equisolid angle mapping function.

2.3.3 The orthographic fisheye model

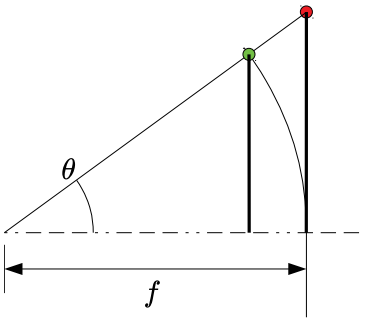


Figure 5: *Orthographic mapping*

The mapping function for this lens type is

$$F_{\text{og}}(\theta) = f_{\text{dn}} \sin\theta \tag{20}$$

The derivative is

$$\frac{d}{d\theta}F_{\text{og}}(\theta) = f_{\text{dn}}\cos\theta \tag{21}$$

The inverse function is

$$F_{\text{og}}^{-1}(r_{\text{dn}}) = \arcsin\frac{r_{\text{dn}}}{f_{\text{dn}}} \tag{22}$$

The derivative of the inverse function is

$$\frac{d}{dr_{\text{dn}}}F_{\text{og}}^{-1}(r_{\text{dn}}) = \frac{1}{\sqrt{f_{\text{dn}}^2 - r_{\text{dn}}^2}} \tag{23}$$

Up to our knowledge this mapping function does not have particular invariance properties. The mapping function simply corresponds to the view of a sphere from a large distance, like e.g. the moon as we see it from earth. If we imagine a sphere with the entire space around it perpendicularly mapped to its surface as a texture, this fisheye mapping shows us (half of) the sphere from a large distance.

2.3.4 The stereographic fisheye model

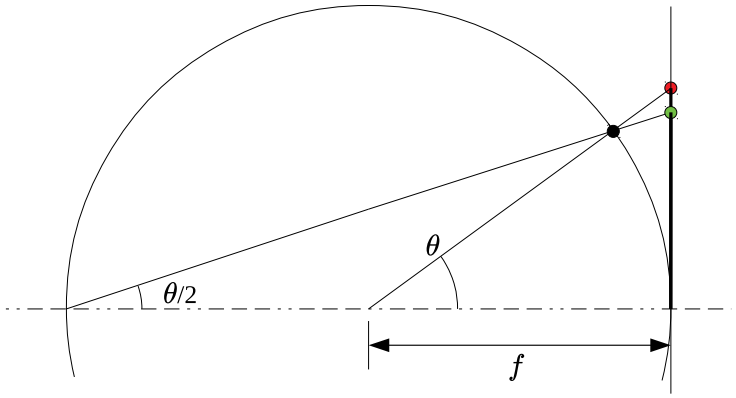


Figure 6: *Stereographic mapping*

The mapping function for this type of lens is

$$F_{\text{sg}}(\theta) = 2f_{\text{dn}}\tan\frac{\theta}{2} \tag{24}$$

The derivative is

$$\frac{d}{d\theta}F_{\text{sg}}(\theta) = f_{\text{dn}}(1+\tan^2\frac{\theta}{2}) = \frac{f_{\text{dn}}}{\cos^2\frac{\theta}{2}}. \tag{25}$$

The inverse function is

$$F_{\text{sg}}^{-1}(r_{\text{dn}}) = 2\arctan\frac{r_{\text{dn}}}{2f_{\text{dn}}} \tag{26}$$

The derivative of the inverse function is

$$\frac{d}{dr_{\text{dn}}}F_{\text{sg}}^{-1}(r_{\text{dn}}) = \frac{1}{f_{\text{dn}}(1+\tan^2\frac{r_{\text{dn}}}{2f_{\text{dn}}})} \tag{27}$$

From a mathematical point of view such a lens is able to map the entire unit sphere around the projection center to a filmback, except for one single point $(0, 0, 1)$; the mapping function becomes singular at $\theta = \pi = 180°$. An example for such a lens is the **Samyang 8mm F3.5** fisheye [3], which has an angle of view of up to 180°.

The rest of this section is somewhat math-heavy. If you are not interested in the invariance properties of the stereographic mapping function, feel free to skip this part.

The stereographic mapping function is conformal, i.e. any angle of (the tangential vectors of) two intersecting curves is preserved under the fisheye mapping. In order to see this we map a scalar product with an appropriate normalization of tangential vectors from filmback to the unit sphere. The proper definition of the scalar product is done by means of a metric tensor. In (diagonally normalized) filmback coordinates, which are polar coordinates in a flat space we have the metric tensor

$$g'(\phi, r_{\mathrm{dn}}) = \begin{bmatrix} r_{\mathrm{dn}}^2 & 0 \\ 0 & 1 \end{bmatrix} \tag{28}$$

while on the unit sphere we have

$$g(\phi, \theta) = \begin{bmatrix} \sin^2\theta & 0 \\ 0 & 1 \end{bmatrix} \tag{29}$$

Let us recall how we map tangential vectors from one manifold to another. Assume we have two manifolds U and V and a differentiable mapping $f : U \to V$. For any point $p \in U$ the tangential space T_pU at p can be defined by means of equivalence classes of differentiable curves in U containing p, where two curves are equivalent iff they have the same derivative at the point p. If we skip the details on charts for coordinate representation we can represent the tangential vector as derivative

$$u = \left.\frac{d}{dt}\right|_0 c(t) \tag{30}$$

where c is any of the curves in the equivalence class mentioned, with $c(0) = p$. The mapped tangential vector $T_pf(u) \in T_{f(p)}V$ is now simply[1]

$$T_pf(u) = \left.\frac{d}{dt}\right|_0 f(c(t)). \tag{31}$$

Our mapping function is restricted to the radial part due to rotational symmetry. As a consequence mapping the ϕ-component (index 0) is the identity, while for the θ-component (index 1) we get some expression. Any tangential vector $a = (a_0, a_1)$

[1]Not sure if T_pf is a common way to write the mapping between the tangent spaces. There seem to be quite a lot of notations.

at some point $p = (\phi, \theta)$ on the unit sphere is mapped to filmback space by

$$\begin{aligned} a_0 \mapsto a'_0 &= a_0 \\ a_1 \mapsto a'_1 &= \left.\frac{d}{dt}\right|_{t=0} F_{\mathrm{sg}}(\theta + t a_1) \\ &= \frac{d}{d\theta} F_{\mathrm{sg}}(\theta) a_1 \\ &= f_{\mathrm{dn}}(1 + \tan^2 \frac{\theta}{2}) a_1 \\ &= \frac{f_{\mathrm{dn}} a_1}{\cos^2 \frac{\theta}{2}} \end{aligned} \tag{32}$$

We consider two tangential vectors $a = (a_0, a_1)$ and $b = (b_0, b_1)$ on the unit sphere at $p = (\phi, \theta)$. We will map these to vectors $a' = (a'_0, a'_1)$ and $b' = (b'_0, b'_1)$ in filmback space and verify that the angle between these vectors is preserved. The angle between the two vectors is

$$\gamma = \arccos \frac{a \cdot b}{|a||b|} \tag{33}$$

Any dot-product occurring here must be expressed by means of the metric tensor g (summation over double indices implied):

$$\begin{aligned} \gamma &= \arccos \frac{g_{\mu\nu}(\phi, \theta) a_\mu b_\nu}{\sqrt{g_{\mu\nu}(\phi, \theta) a_\mu a_\nu} \sqrt{g_{\mu\nu}(\phi, \theta) b_\mu b_\nu}} \\ &= \arccos \frac{a_0 b_0 \sin^2 \theta + a_1 b_1}{\sqrt{a_0^2 \sin^2 \theta + a_1^2} \sqrt{b_0^2 \sin^2 \theta + b_1^2}} \end{aligned} \tag{34}$$

Now we express this by mapped tangential vectors a' and b' as shown in (32).

$$\gamma = \arccos \frac{a'_0 b'_0 \sin^2 \theta + a'_1 b'_1 f_{\mathrm{dn}}^{-2} (\cos^4 \frac{\theta}{2})}{\sqrt{(a'_0)^2 \sin^2 \theta + (a'_1)^2 f_{\mathrm{dn}}^{-2} (\cos^4 \frac{\theta}{2})} \sqrt{(b'_0)^2 \sin^2 \theta + (b'_1)^2 f_{\mathrm{dn}}^{-2} (\cos^4 \frac{\theta}{2})}} \tag{35}$$

We extract the focal length and the $\cos^4$ factor in both numerator and denominator. Using the common relation

$$2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} = \sin \theta \tag{36}$$

allows us to replace

$$\frac{\sin^2 \theta}{\cos^4 \frac{\theta}{2}} = 4 \tan^2 \frac{\theta}{2} \tag{37}$$

and this leads to

$$\gamma = \arccos \frac{a'_0 b'_0 4 f_{\mathrm{dn}}^2 \tan^2 \frac{\theta}{2} + a'_1 b'_1}{\sqrt{(a'_0)^2 4 f_{\mathrm{dn}}^2 \tan^2 \frac{\theta}{2} + (a'_1)^2} \sqrt{(b'_0)^2 4 f_{\mathrm{dn}}^2 \tan^2 \frac{\theta}{2} + (b'_1)^2}} \tag{38}$$

Since due to the mapping function

$$r_{\mathrm{dn}} = 2 f_{\mathrm{dn}} \tan \frac{\theta}{2} \tag{39}$$

we finally get

$$\gamma = \arccos \frac{a'_0 b'_0 r_{\mathrm{dn}}^2 + a'_1 b'_1}{\sqrt{(a'_0)^2 r_{\mathrm{dn}}^2 + (a'_1)^2} \sqrt{(b'_0)^2 r_{\mathrm{dn}}^2 + (b'_1)^2}} \tag{40}$$

which is

$$\begin{aligned} \gamma &= \arccos \frac{g'_{\mu\nu}(\phi, r_{\mathrm{dn}}) a'_\mu b'_\nu}{\sqrt{g'_{\mu\nu}(\phi, r_{\mathrm{dn}}) a'_\mu a'_\nu} \sqrt{g'_{\mu\nu}(\phi, r_{\mathrm{dn}}) b'_\mu b'_\nu}} \\ &= \arccos \frac{a' \cdot b'}{|a'||b'|} \end{aligned} \tag{41}$$

which is the angle of the tangential vectors mapped to the filmback.

2.4 Lens distortion

In order to formulate a lens distortion model we can concatenate the fisheye mapping with some commonly used deformation polynomial. There are in principle three ways to apply the deformation polynomial, either in object space or in fisheye filmback space or even backend in plain filmback space. There are several reasons to define the distortion polynomial in object space, and this is what we do here (see Figure 7)

In the following we shall use a degree-N correction polynomial. Since we assume in our models that their is no azimuthal dependency, we restrict ourselves to the radial component:

$$\begin{aligned} g_{\mathrm{poly,rad}}(\theta) &= \theta(1 + c_2\theta^2 + c_4\theta^4 + \ldots) \\ &= \theta \sum_{i=0,\ \mathrm{even}}^{N} c_i \theta_i \end{aligned} \tag{42}$$

with $c_0 = 1$.

In the current implementation in the LDPK we use a degree-8 polynomial, and we shall define model names for this degree. As usual, the polynomial describes

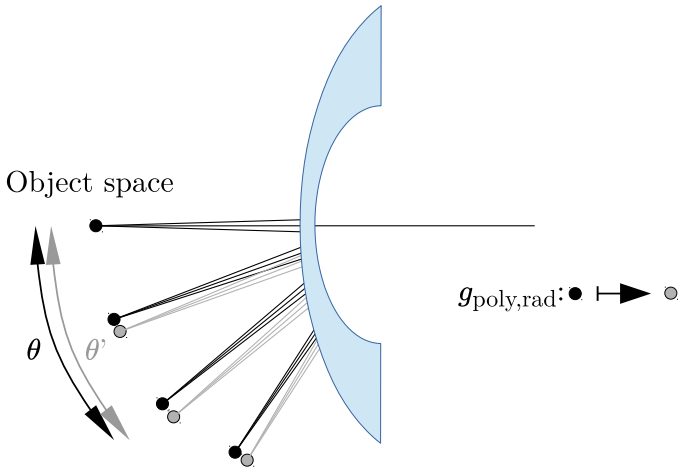


Figure 7: *Removing distortion in object space*

the mapping from a distorted to an undistorted point. The four lens distortion models suggested in this document apply / remove the deformation in angle space, which means the fisheye mapping with a distortion function has the form

$$F \circ g^{-1}_{\text{poly,rad}} \tag{43}$$

Table 1: *Names of distortion models in this document vs.* 3DE4*'s GUI*

GUI	Document
3DE4 Fisheye - Equidistant, Degree 8	Poly-8-Radial-Fisheye-Equidistant
3DE4 Fisheye - Equisolid, Degree 8	Poly-8-Radial-Fisheye-Equisolid
3DE4 Fisheye - Orthographic, Degree 8	Poly-8-Radial-Fisheye-Orthographic
3DE4 Fisheye - Stereographic, Degree 8	Poly-8-Radial-Fisheye-Stereographic

Starting from the remapping function (6) we insert the lens distortion polynomial between the two parts, so that it acts in angular space. The (almost) complete radial distortion model function then reads

$$g_{\text{any,rad}} : r_{\text{dn}} \mapsto r'_{\text{dn}} = G \circ g_{\text{poly,rad}} \circ F^{-1}_{\text{any}}(r_{\text{dn}}) \tag{44}$$

Depending on which of the mapping functions we insert here for F_{any} we get one of the four distortion models listed above.

2.5 Scale

For practical reasons it makes sense to introduce another parameter which acts on the fisheye filmback side, namely the parameter *Scale*. This parameter simply

scales the distorted fisheye image linearly around the lens center. We then have the complete fisheye remapping function in radial coordinates:

$$g_{\text{any,rad}} : r_{\text{dn}} \mapsto r'_{\text{dn}} = G \circ g_{\text{poly,rad}} \circ F^{-1}_{\text{any}} \circ \text{scl}_{s_{\text{uni}}}(r_{\text{dn}}) \tag{45}$$

where scl_s is simply defined as

$$\text{scl}_s : r_{\text{dn}} \mapsto s r_{\text{dn}} \tag{46}$$

Since for the implementation we need this model function as a mapping from (a domain in) $\mathbb{R}^2$ to $\mathbb{R}^2$, we first evaluate the EUCLIDIAN norm and multiply with the unit vector of p_{dn}:

$$g_{\text{any}}(p_{\text{dn}}) \coloneqq \hat{p}_{\text{dn}}\, G \circ g_{\text{poly,rad}} \circ F^{-1}_{\text{any}} \circ \text{scl}_{s_{\text{uni}}}(|p_{\text{dn}}|) \tag{47}$$

The entire list of parameters for all fisheye models is given in Table 2.

Table 2: *Parameters of model* Poly-8-Radial-Fisheye-*

Doc	Code	GUI	Default
c_2	`c2`	*C2 - Degree 2*	0
c_4	`c4`	*C4 - Degree 4*	0
c_6	`c6`	*C6 - Degree 6*	0
c_8	`c8`	*C8 - Degree 8*	0
s_{uni}	`s_uni`	*Scale*	1

2.6 JACOBIAN

The aim of this section is to formulate the JACOBIAN for the model functions. In the math expressions to follow, points p and radii r are always in dn-coordinates, but we omit the index "dn" for better readability. At this point we already know the derivatives of G, $g_{\text{poly,rad}}$ and F^{-1}_{any}, which means we can implement the derivative

$$\frac{d}{dr} g_{\text{any,rad}}(r) = \left.\frac{d}{d\theta'}\right|_{g_{\text{poly,rad}}(F^{-1}_{\text{any}}(r))} G(\theta') \left.\frac{d}{d\theta}\right|_{F^{-1}_{\text{any}}(r)} g_{\text{poly,rad}}(\theta) \frac{d}{dr} F^{-1}_{\text{any}}(r) \tag{48}$$

There is no gain in writing this down explicitly, since in the implementation we simply compute either of the three factors and multiply them, as in the expression above. What we need, however, is the JACOBIAN

$$J_{\mu\nu}(p) = \frac{\partial}{\partial p_\nu} g_{\text{any}}(p)_\mu \tag{49}$$

In order to formulate this we need the partial derivatives

$$\frac{\partial}{\partial p_\nu}\hat{p}_\mu = \frac{1}{|p|}\left(\delta_{\mu\nu} - \hat{p}_\mu\hat{p}_\nu\right) \tag{50}$$

and

$$\frac{\partial}{\partial p_\nu}|p| = \hat{p}_\nu \tag{51}$$

Combining all derivatives we finally get

$$J_{\mu\nu}(p) = \frac{1}{|p|}\left(\delta_{\mu\nu} - \hat{p}_\mu\hat{p}_\nu\right) g_{\text{any,rad}}(|p|) + \hat{p}_\mu\hat{p}_\nu \left.\frac{d}{dr}\right|_{|p|} g_{\text{any,rad}}(r) \tag{52}$$

Again, it does not help to expand this any further, and we implement it in the form given here.

3 Reparametrization

In [2] we have seen how lens distortion parameters are transformed when the filmback position or size is modified. In this section we shall do the same for the four fisheye lens distortion models. At first this may seem to be tricky because lots of trigonometric functions are involved, but in fact the opposite is true: modifying the filmback does *not* affect the lens distortion coefficients. In order to see this let us recall what reparametrization is about. Given a model function, here $g_{\text{any}}[f_{\text{dn}}, c]$ in dn-coordinates, we express this in unit coordinates, which are defined by $(0, 0)$ being the lower left corner and $(1, 1)$ the upper right corner of the field of view (white dotted lines in 3DE4). The relationship between these coordinates is given by an affine mapping ϕ, which depends on the filmback size $w_{\text{fb,cm}}, h_{\text{fb,cm}}$ and the lens center offset $x_{\text{lco,cm}}, y_{\text{lco,cm}}$.

$$(x_{\text{dn}}, y_{\text{dn}}) = \phi(x_{\text{unit}}, y_{\text{unit}}) \tag{53}$$

If we now replace the filmback by a new one, $\tilde{w}_{\text{fb,cm}}, \tilde{h}_{\text{fb,cm}}, \tilde{x}_{\text{lco,cm}}, \tilde{y}_{\text{lco,cm}}$, we get a new mapping $\tilde{\phi}$. The lens distortion parameters c are replaced by new ones $\tilde{c}$, so that the undistorted position with respect to the new filmback is

$$(\tilde{x}'_{\text{unit}}, \tilde{y}'_{\text{unit}}) = \tilde{\phi}^{-1} \circ g_{\text{any}}[\tilde{f}_{\text{dn}}, \tilde{c}] \circ \tilde{\phi}(\tilde{x}_{\text{unit}}, \tilde{y}_{\text{unit}}) \tag{54}$$

We already have shown that

$$\phi(x_{\text{unit}}, y_{\text{unit}}) = \rho\tilde{\phi}(\tilde{x}_{\text{unit}}, \tilde{y}_{\text{unit}}), \tag{55}$$

where ρ is given by the ratio of the diagonal radii of the filmbacks

$$\rho = \frac{\tilde{r}_{\text{fb,cm}}}{r_{\text{fb,cm}}} \tag{56}$$

so that we can conclude

$$g_{\text{any}}[\tilde{f}_{\text{dn}}, \tilde{c}](\tilde{x}_{\text{dn}}, \tilde{y}_{\text{dn}}) = \rho^{-1} g_{\text{any}}[f_{\text{dn}}, c](\rho\tilde{x}_{\text{dn}}, \rho\tilde{y}_{\text{dn}}) \tag{57}$$

The interesting thing about the fisheye models is now that they depend on focal length, which also undergoes the same transform as any radial position in dn-coordinates:

$$\tilde{f}_{\text{dn}} = \frac{f_{\text{cm}}}{\tilde{r}_{\text{fb,cm}}} = \frac{f_{\text{cm}}}{\rho r_{\text{fb,cm}}} = \rho^{-1} f_{\text{dn}} \tag{58}$$

so that finally

$$g_{\text{any}}[\tilde{f}_{\text{dn}}, \tilde{c}](\tilde{x}_{\text{dn}}, \tilde{y}_{\text{dn}}) = \rho^{-1} g_{\text{any}}[\rho\tilde{f}_{\text{dn}}, c](\rho\tilde{x}_{\text{dn}}, \rho\tilde{y}_{\text{dn}}) \tag{59}$$

At this point in [2] we inserted the various model functions and derived the raltionship between c and $\tilde{c}$. Here it is sufficient to consider the radial dependency. The radius r_{dn} exhibits the same behaviour under filmback transformations as the point $(x_{\text{dn}}, y_{\text{dn}})$:

$$r_{\text{dn}} = \rho\tilde{r}_{\text{dn}} \tag{60}$$

Now let us consider the complete remapping function, now with parameters added to our notation

$$\tilde{r}'_{\text{dn}} = G[\tilde{f}_{\text{dn}}] \circ g_{\text{poly,rad}}[\tilde{c}] \circ F^{-1}_{\text{any}}[\tilde{f}_{\text{dn}}](\tilde{r}_{\text{dn}}) \tag{61}$$

The mapping function $F^{-1}_{\text{any}}[\tilde{f}_{\text{dn}}](\tilde{r}_{\text{dn}})$ only depends on the ratio of $\tilde{f}_{\text{dn}}$ and $\tilde{r}_{\text{dn}}$, which means

$$F^{-1}_{\text{any}}[\tilde{f}_{\text{dn}}](\tilde{r}_{\text{dn}}) = F^{-1}_{\text{any}}[f_{\text{dn}}](r_{\text{dn}}) \tag{62}$$

For the remapping function $G[\tilde{f}_{\text{dn}}]$ we have

$$G[\tilde{f}_{\text{dn}}](\cdot) = \rho^{-1} G[f_{\text{dn}}](\cdot) \tag{63}$$

Putting all this together we can only conclude from (59) and (61) that the distortion polynomial remains invariant under the filmback transformation:

$$g_{\text{poly,rad}}[\tilde{c}] = g_{\text{poly,rad}}[c] \tag{64}$$

which means

$$\tilde{c} = c \tag{65}$$

for all fisheye models. This fundamentally different behaviour as opposed to the standard models in [2] results from the fact that we normalize the input r_{dn} by the focal length f_{dn}. We could have constructed our standard models in the same way, but this has the disadvandatge that then the entire distortion behaviour depends on focal length. For the fisheye lenses we cannot avoid this, but for the plain lenses it is not desireable.

4 List of symbols

B_{any}	Domain of a fisheye mapping function
$\bullet_{\text{dn}}$	Index indicating diagonally normalized coordinates
f_{dn}	Focal length in diagonally normalized coordinates
F_{ed}	Equidistant mapping function, preserves distances from center
F_{es}	Equisolid mapping function, preserves solid angles
F_{og}	Orthographic mapping function
F_{sg}	Stereographic mapping function, preserves angles
F_{any}	Placeholder for any of F_{ed}, F_{es}, F_{og}, F_{sg}.
$g_{\mu\nu}(\dots)$	Metric tensor
$g_{\text{poly,rad}}$	Distortion polynomial in radial direction
g_{any}	Fisheye distortion model function
$g_{\text{any,rad}}$	Fisheye distortion model function in radial direction
G	Remapping from unit sphere to plain filmback space
$J_{\mu\nu}(\dots)$	Jacobian
p_{dn}	A point in fisheye filmback space
p'_{dn}	A point in plain filmback space
r_{dn}	Distance of a point from the lens center in fisheye filmback space
r'_{dn}	Distance of a point from the lens center in plain filmback space
ϕ	Azimuth angle in both object space and filmback space
θ	Polar angle in object space, between object point and axis of view
T_pU	Tangential space at p of a manifold U
T_pf	Tangential lift of a function $f : U \to \dots$ to tangent bundle TU at p
$x_{\text{obj}}, y_{\text{obj}}, z_{\text{obj}}$	Coordinates in object space

References

[1] Creative Commons Attribution-ShareAlike 3.0
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[2] U. Sassenberg, The Standard Models for Lens Distortion in 3DE4, `$LDPK/doc/tex/standard/tde4_ldm_standard.pdf`, 2018

[3] www.samyanglens.com

[4] paulbourke.net/dome/fisheyetypes/

[5] http://www.fischaugenobjektiv.de/html/fisheyeMath.html